

PROJECT B · CUFFLESS BLOOD PRESSURE

Model training — round one

Three of five models, each tested four ways, plus the external test
on the wearable dataset

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Every number generated from the saved run files — [results/overview/RESULTS.md](#)

Three models done, two to go

The order is the spec's, section I.

	Model	What it reads	Status
1	Random Forest	61 measurements taken from each clip	done
2	Gradient Boosting	the same 61 measurements	done
3	1D-CNN	the raw waveform, 2×1250 numbers	done
4	ResNet1D	the raw waveform	next
5	Transformer	the waveform + age, sex, BMI	after that

15 training runs. Each learns from 419,040 clips from 1,164 patients.

The first two never see the signal — only measurements taken from it beforehand.

The same model, tested four ways

Nothing changes between these except who is in the test set.

Test rule	Who is tested	Met them before?
leaky	same patients, clips picked at random	yes
calbased	same patients, official held-out clips	yes
calfree	144 patients	no
aami	116 patients, chosen to span the whole BP range	no

calfree is the honest one — the situation a real user is in: the device has never met them.

THE TWO CLINICAL STANDARDS USED IN THE TABLES THAT FOLLOW

- AAMI

pass or fail, and it never looks at MAE. The bias must be within ± 5 mmHg and the spread of the error within 8. Both, or it fails.
- BHS

a grade from A to D on how many predictions land close: grade A needs 60% of them within 5 mmHg, 85% within 10, and 95% within 15.

All four test rules

Mean absolute error in mmHg, lower is better. The two clinical standards are shown for diastolic: systolic fails AAMI in every row, so it would carry no information.

Test rule	Who is tested	SBP	DBP	AAMI (on DBP)	BHS (on DBP)
leaky	same patients, clips picked at random	7.47	4.34	PASS	A
calbased	same patients, official held-out clips	7.14	4.11	PASS	A
calfree (A)	144 patients it has never seen	12.76	8.38	FAIL	D
aami	116 new patients, full BP range	19.67	12.24	FAIL	D
calfree (B)	the same 144, without BP-bin weighting	12.93	8.45	FAIL	D

(A) carries the BP-bin weighting the spec asks for. (B) is the same model without it — the form every published number takes.

Known patients 7.31 → new patients 12.76 gap 1.75x

Best-looking model on the two rules where it already knows the patient, and no better than the others once the patients are new.

All four test rules

Mean absolute error in mmHg, lower is better. The two clinical standards are shown for diastolic: systolic fails AAMI in every row, so it would carry no information.

Test rule	Who is tested	SBP	DBP	AAMI (on DBP)	BHS (on DBP)
leaky	same patients, clips picked at random	10.29	6.13	PASS	B
calbased	same patients, official held-out clips	10.20	6.03	PASS	B
calfree (A)	144 patients it has never seen	14.63	9.00	FAIL	D
aami	116 new patients, full BP range	17.44	11.14	FAIL	D
calfree (B)	the same 144, without BP-bin weighting	12.71	8.25	FAIL	D

(A) carries the BP-bin weighting the spec asks for. (B) is the same model without it — the form every published number takes.

Known patients 10.24 → new patients 14.63 gap 1.43x

Steadier than the forest — worse on known patients, better on new ones, so its gap is smaller.

All four test rules

Mean absolute error in mmHg, lower is better. The two clinical standards are shown for diastolic: systolic fails AAMI in every row, so it would carry no information.

Test rule	Who is tested	SBP	DBP	AAMI (on DBP)	BHS (on DBP)
leaky	same patients, clips picked at random	10.76	6.88	FAIL	C
calbased	same patients, official held-out clips	12.84	8.04	FAIL	D
calfree (A)	144 patients it has never seen	14.86	9.01	FAIL	D
aami	116 new patients, full BP range	17.33	11.77	FAIL	D
calfree (B)	the same 144, without BP-bin weighting	12.07	7.98	FAIL	D

(A) carries the BP-bin weighting the spec asks for. (B) is the same model without it — the form every published number takes.

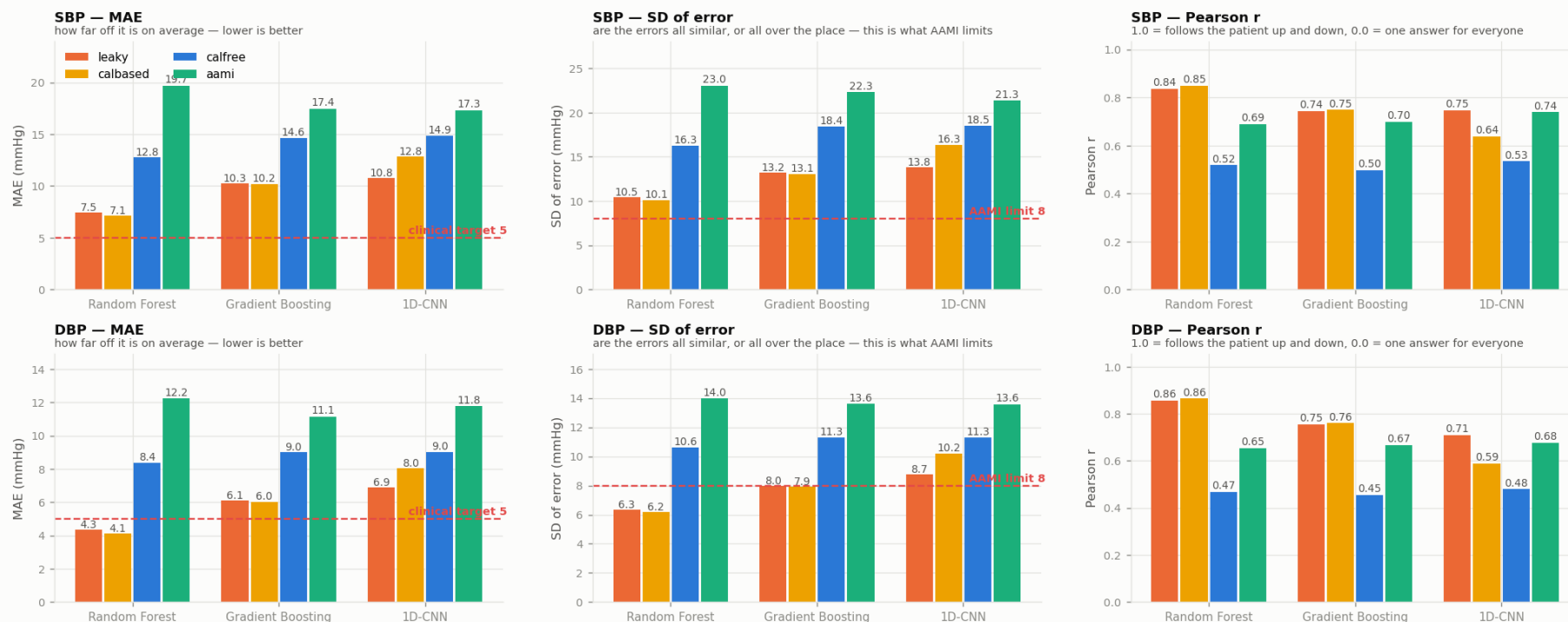
Known patients 11.80 → new patients 14.86 gap 1.26x

The only one that reads the waveform itself. 381,186 parameters, 20 epochs, about 30 minutes on a T4 GPU.

All three models, all four test rules

Top row systolic, bottom row diastolic. Three different questions, left to right.

Same model, same training clips — only the test patients change. Orange = the model had already met them; blue and green = it had not.



MAE — how far off it is on average. Lower is better. This is the number papers headline.

SD of error — whether the errors are all similar or all over the place. This is what AAMI limits, not MAE.

Pearson r — 1.0 means it follows the patient up and down; 0.0 means one fixed answer for everyone.

The red dotted line is the clinical limit — bars should sit below it. None of them do.

The gap is not the same for every model

Model	Known patients	New patients	Gap
Random Forest	7.31	12.76	1.75x
Gradient Boosting	10.24	14.63	1.43x
1D-CNN	11.80	14.86	1.26x

The Random Forest memorises the patient, so a new clip from a known patient is still easy — it looks excellent under the first two rules.

The CNN memorises the clip, which does not help even on the same patient. Both are overfitting — but the leaky rule only rewards the first kind.

What it is, and where it comes from

The proposal asks for it by name, in section L:

"Class/target imbalance handling: BP value distribution is typically right-skewed in ICU populations (more hypertensive/hypotensive extremes); apply stratified sampling or a weighted loss over BP value bins."

The problem it solves:

Systolic band	Training clips	Share of the data	Weight each clip is given
below 90	33,580	7.2%	2.3x
90 – 130	333,610	71.7%	0.5x
130 – 170	95,152	20.5%	1.0 – 4.4x
above 170	3,138	0.7%	24.7x

Seven in ten training clips sit between 90 and 130, so a model that always answers "about 115" is right most of the time — and useless for the patients a cuffless monitor would exist to help.

Balancing makes one clip above 170 count as much as twenty-five ordinary ones.

What it costs and what it buys

Systolic MAE on the 144 unseen patients, with the weighting off and on.

Model	Overall off	Overall on	Change	Patients above 170 off	Patients above 170 on	Change
Random Forest	12.93	12.76	-0.17	47.18	43.13	-4.05
Gradient Boosting	12.71	14.63	+1.92	46.21	32.93	-13.28
1D-CNN	12.07	14.86	+2.79	47.10	37.62	-9.48

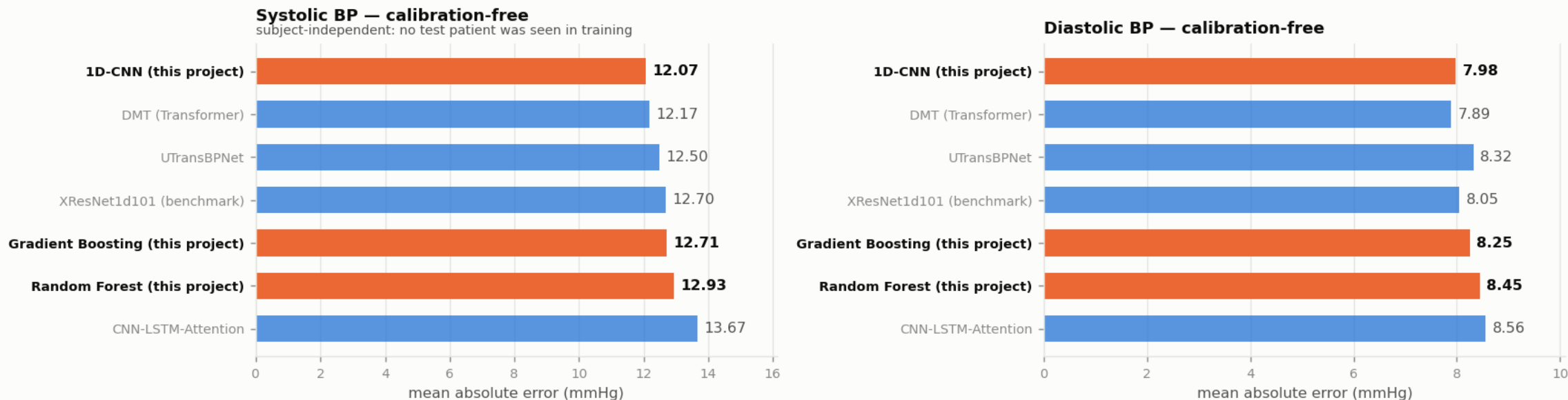
Read the two "Change" columns against each other. On gradient boosting the weighting costs 1.92 mmHg on the average and buys 13.29 on the patients above 170. On the CNN it costs more. On the random forest it costs nothing at all — the average improves as well.

So both numbers are reported everywhere in this deck:

- (A) with the weighting — this project's model, what section L asks for
- (B) without it — how every published number was measured, so the comparison holds

Same dataset, same official splits, same metric

Published results on PulseDB, honest protocol — orange is this project, measured without BP-bin weighting so the comparison holds



Orange is this project. A four-layer CNN lands at 12.07 mmHg; a 2026 transformer is at 12.17.

That is not a win — it is evidence that the architecture is not what is holding the field back.

What PPG-DaLiA actually is

15 volunteers, a wrist watch and a chest strap, one ordinary day each.

One volunteer's day — S5, age 21:

transient	43 min
lunch	38 min
working	21 min
driving	14 min
sitting	10 min
walking	10 min
stairs	8 min
cycling	7 min
table soccer	5 min

What the researchers recorded:

wrist PPG	✓
chest ECG	✓
accelerometer	✓
heart rate	✓
what they were doing	✓
age, sex, height, weight	✓
BLOOD PRESSURE	✗ never measured

Nobody ever put a blood-pressure cuff on these fifteen people. The dataset was built to study heart rate under motion, not blood pressure.

That is why the next slide has no accuracy number for BP — there is nothing to compare against.

Does it work on a watch?

Trained on hospital patients, then run on 15 volunteers wearing a wrist watch.

There is no MAE for blood pressure here, and there cannot be: MAE needs a true value to subtract from, and PPG-DaLiA never measured anyone's blood pressure. Two things it does have:

1. HEART RATE — this one is labelled, so it has a real MAE

	MAE (bpm)
volunteer sitting still	13.06
volunteer moving	33.35
all 63,180 clips	27.11

2. BLOOD PRESSURE — no labels, so only where it lands

	Average SBP	Which band
a healthy adult in their 20s	under 120	normal
Random Forest says	130.7	stage 1 hypertension
Gradient Boosting says	135.0	stage 1 hypertension
1D-CNN says	127.0	elevated

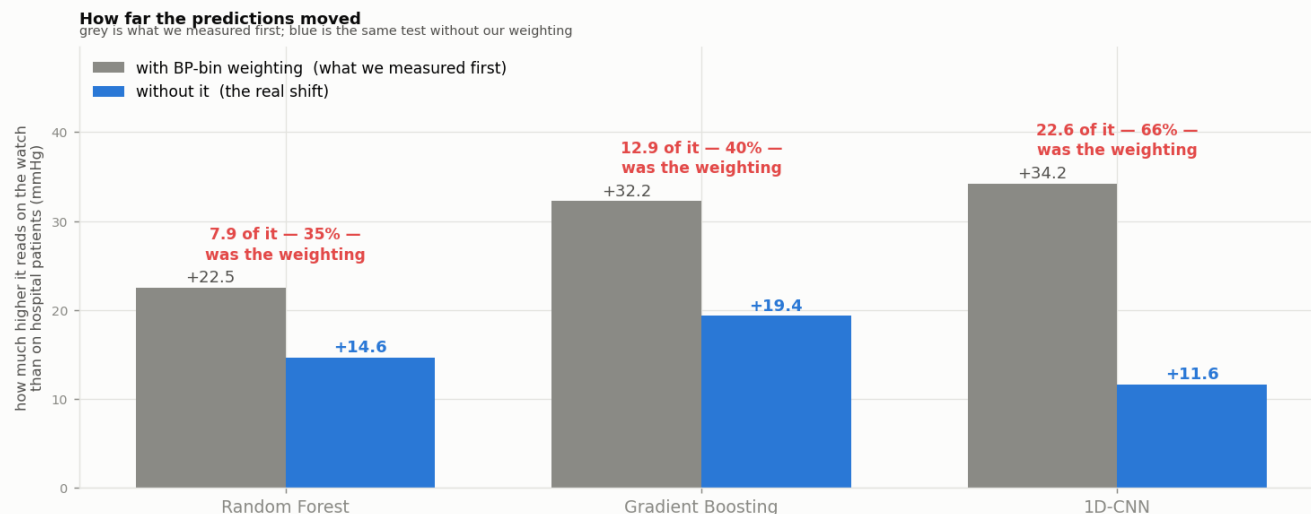
Heart rate is the honest measure, and it is bad: 13 bpm sitting still, 33 moving. A published paper doing this properly on the same data reports 8.69 bpm.

The first row on the right is the clinical norm, not our data — every model lands above it.

Was the shift real, or did we cause it?

The same test, run twice per model — once with the BP-bin weighting, once without.

Note what this measures. It is our model's average on PPG-DaLiA minus our model's average on PulseDB — both are our own outputs, so no BP label is needed anywhere. It is a shift, not an error.

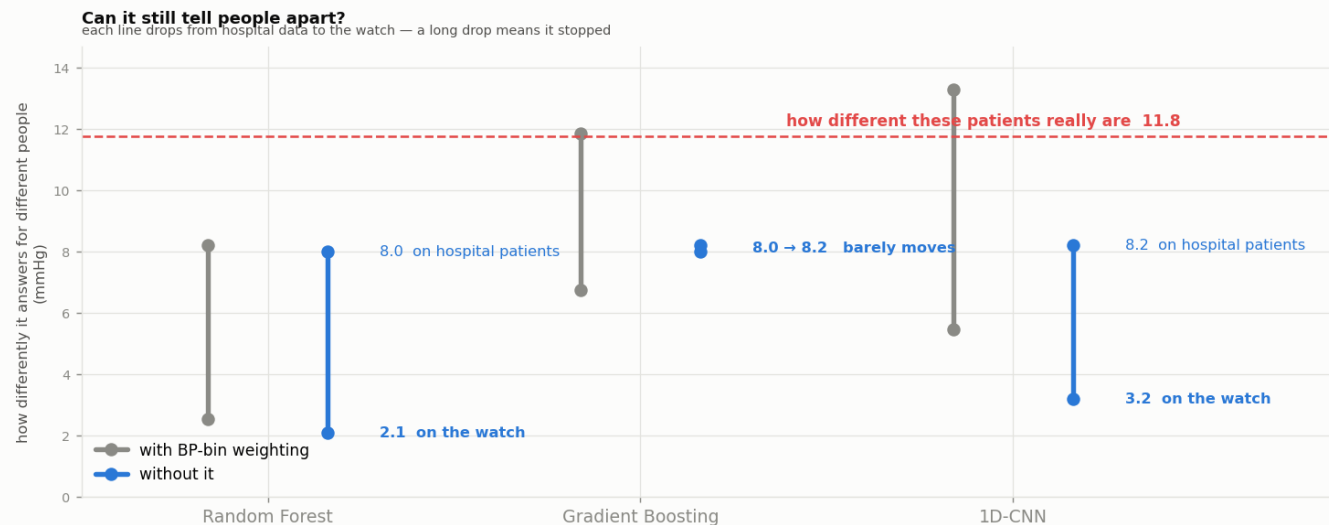


Take the Random Forest. We first measured +22.5. Without the weighting the same test gives +14.6. So 7.9 of that 22.5 — 35% — was our own weighting, and only 14.6 was the watch. On the CNN it was 22.6 out of 34.2: two thirds of what we first called domain shift was us.

Can it still tell people apart?

A separate question — and here the weighting turns out not to be the explanation.

These patients really do differ from one another by about 11.8 mmHg. A model that arrives at the watch still answering that differently is distinguishing people; one answering 2 is not.



The Random Forest goes from 8.0 on hospital patients to 2.1 on the watch — it has started giving all fifteen volunteers nearly the same number. The CNN drops to 3.2. Gradient boosting barely moves. Grey and blue sit on top of each other here, so unlike the shift, this collapse is not our doing.

What comes after this

	Work	Spec
1	ResNet1D and the Transformer — the two remaining models	Month 4
2	Few-shot personalization: give the model a few labelled readings from a new person, then re-measure	Months 6-7
3	How many readings are needed — 1, 3 or 5	Month 9

One question for you:
Section G says PPG-DaLiA "does not contain BP labels at all", but section E asks how much personalization helps there. Calibration needs labels — should this be the distribution check only?

Everything above is in results/overview/RESULTS.md, and any row reproduces with one command: